

Summary

Project Title (Week-1) : House Price Prediction

Name: Syed Uzma Farzeen

Objective

The main aim of this project was to predict house prices using machine learning. I used different features such as area, bedrooms, bathrooms, stories, parking, and other house facilities to understand how they affect the price of a house.

Data Preparation

I loaded the housing dataset and explored it to understand its structure. I checked for missing values and duplicate records, but the dataset did not contain any missing or duplicate data. I then converted the categorical columns into numerical values so that the machine learning models could process the data.

Model Building

I trained two machine learning models: **Linear Regression** and **Random Forest Regressor**. The dataset was split into **80% training data** and **20% testing data**. I evaluated both models using **MAE**, **RMSE**, and **R² Score** to compare their performance.

Findings

From the correlation heatmap and feature importance chart, I found that **area** has the biggest influence on house prices. **Bathrooms**, **air conditioning**, **parking**, and **stories** also play an important role in determining the price. The histogram showed that most houses are in the medium price range, while only a few houses have very high prices.

Model Performance

After comparing both models, I found that **Linear Regression performed slightly better than Random Forest** on this dataset. It achieved an **R² score of 0.653**, which means it could explain about **65% of the variation** in house prices. It also had lower MAE and RMSE values, making it the more accurate model for this dataset.

Conclusion

This project helped me understand how machine learning can be used to predict house prices. I learned that the size of the house and important facilities have a strong impact on its price. I think real estate companies can use these models as a supporting tool to estimate house prices more accurately and make better pricing decisions.